VISHWATEJ KHOT

Forward Deployed Engineer | AI-Native Builder, Client-Embedded

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PROFESSIONAL SUMMARY

AI Engineer with 2.5 years of experience across UK-based companies, embedding directly with clients to understand real workflows and building production AI systems myself, not scoping and handing off. At Lume Studios I integrate AI directly into my own development workflow and build production systems for enterprise clients end-to-end, then build the evaluation infrastructure that proves a system holds up, not just that it demos well. I write code that works and that others can follow, and I walk into ambiguous problems with no spec and work out what to build, evidenced repeatedly across every engagement I've owned. I have hands-on experience with LLM integration and prompt engineering in production, not just prototypes, and I communicate technical trade-offs clearly to non-technical clients. As Founder of Vishverse I built and shipped a full AI platform solo, entrepreneurial and comfortable in ambiguity by necessity. I hold an MSc in Data Science from the University of Bristol. Published researcher (ICETC 2025).

TECHNICAL SKILLS

Client-Embedded Building: Understanding real workflows and constraints beyond stated requirements; building the solution personally, not scoping and handing off; post-launch monitoring and debugging in client environments

AI-Assisted Development: Highly fluent with Claude Code for daily development, scaffolding, and debugging; ships faster without sacrificing quality; hand-writes core logic and prompt design

LLM Integration & Prompt Engineering: Production LLM integration (not just prototypes): LangChain, LangGraph, LlamaIndex, MCP, A2A protocol, prompt engineering, ReAct agents, multi-agent orchestration

Evaluation Pipelines: Evaluation frameworks gating production release, confidence scoring, structured audit logging, systematic monitoring of deployed system performance

Full-Stack & Data Plumbing: Python, TypeScript, FastAPI, Flask, REST APIs, data pipelines, structured extraction, PostgreSQL (Supabase with pgvector + RLS), Docker, CI/CD, AWS

Ambiguity, Proactivity & Security Awareness: Walks into a messy problem with no spec and works out what to build; asks the right questions early to find the real problem; treats secure-by-design as part of delivery, not an afterthought

WORK EXPERIENCE

Forward Deployed Engineer | *Lume Studios Consulting, United Kingdom* Jan 2026 – Present

- **Embed with client teams to understand real workflows, systems, and problems**, not just stated requirements: architected a multi-client RAG pipeline (parse, contextual enrichment, embedding, hybrid retrieval, reranking) exposed as an MCP server, deployed via Docker.
- **Build production LLM systems across the full stack, myself**: integration, prompt engineering, evaluation pipelines, and APIs, with no handoff to another team.
- **Monitor and maintain deployed systems post-launch**: built an evaluation framework benchmarking chunk quality, reranking precision, and answer groundedness that gates every client go-live, then debug issues as they arise in production.
- **Asked the right questions early to find the real problem**: identified a client's broken Excel workflow beyond the original scope and shipped an LLM extraction pipeline, cutting manual retrieval time by 60%.

Founder & AI Engineer | *Vishverse, United Kingdom* Oct 2025 – Present

- **Walked into total ambiguity and built the answer myself**: built and shipped Aria, a production multi-tenant AI platform for hospitality businesses across UK, India, Dubai, and USA, solo.
- **Built for maintainability, not just speed**: LLM gateway with Claude, Groq, and OpenAI in a fallback chain, circuit breaker, semantic caching, and daily spend caps, using **MCP and A2A protocols**.
- **Owned data plumbing and security-aware design**: Flask REST API with 40+ endpoints, Supabase multi-tenant architecture with row-level security, deployed on AWS with Docker and CI/CD.

AI Engineer | *Abhinaya Ltd, United Kingdom* Nov 2024 – Aug 2025

- Shipped DocAssist, a production healthcare app (LLMs + RAG) for automated medical documentation and diagnostic decision support, reducing documentation time by 50%, with **evaluation passes run before anything shipped**.
- Built LangChain summarisation engines and vector-DB knowledge modules; maintained FastAPI endpoints serving LLM responses to downstream systems in production.

Generative AI Intern | *Intelimek Systems Pvt Ltd, Pune* Apr 2024 – Mar 2025

- Engineered a production RAG system (LangChain + LLaMA) for fast, accurate retrieval across large PDF corpora, cutting manual processing effort by 40%.

AI Intern | *Abhinaya Ltd, United Kingdom* May 2024 – Nov 2024

- Developed NeRF and Gaussian Splatting pipelines for 3D scene reconstruction, improving volumetric rendering performance by 30% — **picked up an unfamiliar domain fast**, with no prior spec to follow.

PROJECTS

Lloyds Banking Group SME Intelligence Platform | MSc Dissertation, University of Bristol

- **Delivered directly for Lloyds Banking Group's Business and Commercial Banking division** as technical lead of a 4-person MSc dissertation team: built an SME intelligence platform processing **3.1 million UK companies** from Companies House registry data to predict Growth Opportunity, Risk Signal, and Lending Need.
- **Caught and corrected target label leakage** that would have inflated results, rejecting a good-looking result in favour of leakage-safe, defensible model performance.

MedAssist AI: NHS-Aligned Clinical Triage and RAG System

- Hybrid RAG with FAISS, BM25, and RRF fusion, including live confidence scoring and a contradiction-aware ReAct agent for NICE guideline conflicts, escalating to a human when confidence is low. Production stack includes an XGBoost triage classifier, rate limiting, and structured audit logging.

FinSight: Agentic Financial Due-Diligence System

- 4-agent LangChain pipeline (Planner, Retriever, Contradiction, Memo) performing end-to-end due diligence across Lloyds filings, with every claim traced to its source; outputs a structured investment memo via a Streamlit UI with live FAISS index sync.

AegisAI: Self-Healing Enterprise RAG Chatbot

- Production RAG chatbot with confidence scoring, safety enforcement, and non-blocking recovery workflows, **reducing hallucination failures by 40%**. Delivered with a Flask API, real-time monitoring dashboards, and full audit logs — **monitored and maintained post-launch**.

Multiclass Malware Detection: Modified GNN and Explainable AI | Published ICETC 2025

- Built scalable feature pipelines converting raw BYTE files to image representations for ML training, with Explainable AI reporting that reduced analyst investigation time by 45%.

EDUCATION

MSc Data Science | *University of Bristol* Sep 2025 – Sep 2026

B.E. Computer Engineering (CGPA 8.69) | *SPPU University, Pune* May 2021 – May 2025

Diploma in Data Science | *IIT Madras (Remote)* Jan 2021 – Sep 2024

CERTIFICATIONS AND ACHIEVEMENTS

- Research published at ICETC 2025: Multiclass Malware Detection using Modified GNN and Explainable AI. DOI: 10.17492/jpi.pcet.022501
- IIT Madras ML Competition: 5th place nationally. Smart India Hackathon: 8th place at college level.
- Certifications: Machine Learning (Microsoft) · Generative AI (Microsoft) · Data Analysis (Google) · Cloud Computing (AWS)